

Jonathan Carroll, PhD

Quantitative Scientist & Research Software Engineer

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Profile

Quantitative scientist with over a decade of experience applying statistical modelling, data analysis, and research software development across biomedical research, clinical trials, and fisheries science. Author of *Beyond Spreadsheets with R* (Manning Publications, 2018). Specialist in R with production experience across the full analytical pipeline; from data ingestion and curation through modelling, visualisation, and reporting to non-technical audiences.

Extensive experience with clinical trial data standards (ADaM, SDTM), multi-modal biomedical data integration, and survival and longitudinal analyses. Background in translational immunology, genomics, and large-scale epidemiological cohort data (UK Biobank). Comfortable operating independently or leading distributed teams in remote environments since 2017.

Book: *Beyond Spreadsheets with R* (Manning Publications, 2018, 352 pp.) A beginner's guide to R and RStudio
<https://www.manning.com/books/beyond-spreadsheets-with-r>

Technical Skills

Statistical methods	GLMs, GAMs, mixed effects models, survival analysis (Kaplan–Meier, Cox PH), longitudinal data analysis, spatiotemporal modelling, DGE/GSEA, likelihood-based inference
Primary language	R (base and tidyverse, ggplot2, Bioconductor, shiny, targets, dplyr, data.table, Quarto/R Markdown)
Additional languages	Python, Rust, Haskell, Julia, SQL, Cypher, Bash, C++, JavaScript, APL, Lisp, Scala
Data & infrastructure	DuckDB, PostgreSQL, SQLite, Redshift, REST APIs, Docker, Git, CI/CD, UNIX/Linux
Clinical data	ADaM, SDTM, CDISC; FAIR data principles; MultiAssayExperiment
Bioinformatics	Benchling, Sapio LIMS, UK Biobank, genomics pipelines, antibody discovery infrastructure
Reporting	R Markdown / Quarto, shiny dashboards, peer-reviewed publications, policy briefings, conference presentations

Industry Experience

Senior Research Software Engineer

2022 – May 2026

Human Immunology Biosciences (HIBio) & Biogen

- Built analytical infrastructure and exploratory dashboards supporting translational science and antibody discovery; connected data across internal systems via REST APIs.
- Migrated a production LIMS from Benchling to Sapio, preserving data integrity and provenance across complex experimental workflows.
- Processed and retrieved petabytes of UK Biobank genotype data for allele-frequency analyses; productionised and extended C++ likelihood code (TrypLik), orchestrating parallel execution via Bash.
- Managed deployment of GoAnywhere secure file transfer for regulated data exchange.

Research Software Engineer (Contract)

2024 – Dec 2025

Rancho Biosciences / Roche / Genentech

- Integrated AI/LLM tooling into R shiny dashboards for exploratory clinical data analysis; prompt engineering for biomedical summarisation tasks.
- Data processing and analysis in Redshift; exploration of ADaM and SDTM clinical trial datasets.

Technical Lead / co-Product Owner

2021 – 2022

Cancer Immunology, Genentech / Roche

- Led development of a large-scale interactive genomics visualisation platform (R shiny) for life scientists without coding experience; coordinated a distributed team of 10 developers across multiple time zones.
- Platform incorporated DGE, GSEA, Kaplan–Meier survival analyses, volcano plots, t-tests, and laboratory result trend visualisations for clinical trial cohorts.
- Managed agile delivery, CI/CD pipelines, database design (SQLite/SQL), and Docker-based deployment.

Data Curator / Integrator & Software Developer

2017 – 2022

Cancer Immunology, Genentech / Roche

- Curated and integrated multi-modal high-dimensional data (genomics, proteomics, clinical trial outcomes) from ADaM/SDTM-structured datasets into a FAIRified internal DataMart (MultiAssayExperiment).
- Performed Kaplan–Meier and Cox proportional hazards survival analyses for clinical trial cohorts; longitudinal modelling of concomitant medications and laboratory results across trial timepoints.
- Conducted in silico analyses linking multi-omics data to patient outcomes; contributed analyses to peer-reviewed publications (Nature Communications, PNAS, medRxiv).
- Developed R/Bioconductor tooling and shiny applications; authored reusable packages supporting downstream analytical teams.

Principal Consultant

2015 – present

Irregularly Scheduled Programming (ABN: 43 486 039 320)

- Independent statistical and data science consulting; selected clients include Gate Biosciences and Lander Analytics.
- Engagements covering R-based analytical pipelines, data infrastructure, visualisation, and reporting.

Senior Research Officer / Fisheries Modeller

2013 – 2017

South Australian Research and Development Institute (SARDI)

- Developed and maintained stock assessment models for South Australian fisheries using GLMs, GAMs, mixed effects models, and custom likelihood-based approaches in C++ (ADMB).
- Conducted spatiotemporal catch rate analyses and GIS analyses to characterise population dynamics under environmental covariates (wind, swell).
- Produced annual statistical reports and peer-reviewed publications; provided statistically rigorous briefings with uncertainty quantification to government policymakers to inform quota-setting decisions.
- Bridged the gap between scientific evidence and fisheries management, translating complex model outputs — including confidence intervals and likelihood profiles — for non-technical audiences including ministers and industry stakeholders.

Research Experience

Postdoctoral Research Fellow

2009 – 2013

Special Research Centre for the Subatomic Structure of Matter, University of Adelaide

- Dense matter nuclear astrophysics: equation-of-state modelling for hybrid neutron stars.
- Muonic hydrogen spectroscopy and the proton radius puzzle; nonperturbative relativistic quantum mechanical calculations.
- Numerical simulation in Fortran with MPI parallelisation; statistical analysis of spectroscopic data.

Education

PhD, Theoretical Physics

2006 – 2009

University of Adelaide

Thesis: *Applications of the Octet Baryon Quark-Meson Coupling Model to Hybrid Stars* — arXiv:1001.4318

Bachelor of Science (Honours), Physics & Theoretical Physics

2002 – 2005

University of Adelaide

Selected Publications

Book

Carroll J. (2018). *Beyond Spreadsheets with R*. Manning Publications, 352 pp. manning.com/books/beyond-spreadsheets-with-r

Peer-reviewed articles (selected)

Negative impact of the GABA pathway on PD-1/PD-L1 immunotherapy. *Submitted 2025*.

Natural killer cell educating KIR/HLA combinations impact survival in anti-PD-L1 treated cancer patients. *medRxiv*, 2022.

Genetic variation associated with thyroid autoimmunity shapes the systemic immune response to PD-1 checkpoint blockade. *Nature Communications* 12, 3355 (2021).

Polygenic risk for skin autoimmunity impacts immune checkpoint blockade in bladder cancer. *PNAS* 117(22), 12288–

12294 (2020).

Influence of wind and swell on catch rates in a dive fishery: a case study from the South Australian abalone fishery. *Journal of Shellfish Research* 35(3), 685–694 (2016).

Setting quotas using provisional data: a case study from the South Australian abalone fisheries. *New Zealand Journal of Marine and Freshwater Research* 50(3), 371–388 (2016).

Spawning biomass of jack mackerel off eastern Australia. *Fisheries Research* 179, 10–22 (2016).

Nonperturbative relativistic calculation of the muonic hydrogen spectrum. *Physical Review A* 84(1) (2011).

Selected Open-Source Software

R packages on CRAN / Bioconductor (author & maintainer unless noted): {ggeasy} · {charcuterie} · {ntfy} · {mathpix} · {DFplyr} (Bioconductor) · {benchlingapi} · {weatherOz} (contributor) · {datapasta} (contributor)

Community & Communication

- **Blog:** Regular technical writing at jcarroll.com.au covering R, data analysis, and multi-language programming explorations (R, Rust, Haskell, Julia, APL, and others).
- **rweekly.org:** Editor — compilation, curation, and editing of the weekly R community newsletter (2016 – present).
- **Conference presentations:** DataEngBytes and other data science venues.
- **Chess Club Teacher** (volunteer): Highbury Primary School (2022 – present).
- **Governing Council:** Highbury Preschool / Highbury Primary / Modbury High (Chair, Vice Chair; 2016 – present).

Last updated: July 2026